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Enantiodiscrimination of carboxylic acids using the diphenylprolinol NMR chiral solvating agents†

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Enantiopure diphenylprolinols were synthesized from a commercially available starting material. The utility of ¹H NMR spectroscopy for the differentiation of enantiomers using these chiral compounds as CSAs is stated, and their capacity acting as receptors for various carboxylic acids *via* hydrogen bonding is exploited. A linear relationship has been observed between the experimental and observed values of ee indicating the possible use of these compounds for quick and reliable analysis of enantiomerically enriched samples of various mandelic acids. From the experiments performed a preliminary conclusion indicated that the diphenylprolinol **1b** with the free NH and OH is most effective in the chiral discrimination of carboxylic acids in ¹H NMR.

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Chiral molecular recognition is an essential phenomenon involved in a wide variety of areas, ranging from chiral separations and analysis, asymmetric catalysis, to biological and host-guest chemistry.^{1,2} However, among the large numbers of analytical techniques, NMR spectroscopy has proven to be a powerful and versatile tool for the discrimination of chiral molecules and the precise measurement of enantiomeric contents in the last few decades.³ And, if all things go well, micro-coil and microchip NMR are becoming more widely available with the rapid development of miniaturization of NMR systems in the near future.⁴ NMR spectra of enantiomers are identical in achiral medium, and usually have no difference between the spectra of a racemic mixture and that of a single isomer. However, in the classical approach, enantiodiscrimination is achieved by converting substrates to diastereomers by utilizing an enantiopure chiral solvating agent (CSA), chiral lanthanide shift reagent (CLSR), chiral derivatizing agent (CDA) and/or by aligning in chiral anisotropic medium such as PBLG liquid crystal.⁵ In a rare but ideal case, a catalytic amount of reagent is enough for chiral discrimination in solution NMR. Therefore, the use of CSAs for enantiodiscrimination can be attractive as there is no need for derivatization and/or further purification step as in CDA, no paramagnetic signal broadening as in LSR appears, and no isotopical labeling as in chiral alignment media is required. Meanwhile, the

CSAs can differentiate enantiomers due to the formation of diastereomeric complexes held together by weak intermolecular interactions such as van der Waals forces and/or hydrogen bonds following a large enough chemical shift non-equivalence to give clear baseline separation of the NMR signals. Therefore, the CSA is usually a better alternative for enantiodiscrimination owing to its fast and simple approach.⁶

Chiral carboxylic acids and their derivatives are organic molecules involved in a wide variety of biological processes and the investigation of the chiral recognition of carboxylic acids by artificial receptors was of critical importance in the preparation, separation, and analysis of enantiomerically pure carboxylic acids and disclosing the mechanism of interaction of the carboxylic acids with biological systems.⁷ The growing use of optically pure carboxylic acids has given rise to the need for the development of fast and accurate methodologies for the determination of the enantiomeric composition of chiral carboxylic acids. In the past few decades, a variety of chiral NMR solvating agents has been the focus of extensive investigation to determine the enantiomeric purity and understand the basic mechanism of host-guest complexation, particularly for carboxylic acids, such as chiral amines,⁸ amino alcohols,⁹ macrocyclic amines and amides,¹⁰ calixarenes,¹¹ crown or azacrown ethers,¹² BINOL and their derivatives,¹³ chiral thiourea moieties,¹⁴ *etc.* To date, however, there are only a few examples of optically active proline-derived receptors for the enantiomeric recognition of carboxylic acids.¹⁵ Therefore, the development of structurally simple yet efficient receptors for the enantioselective recognition of carboxylic acids still remains a challenging goal.

Among the variety of commercially available amino alcohol derivatives well documented in the literature, proline-derived diphenylprolinols have usually been widely employed as chiral

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ligands in the field of asymmetric synthesis and catalysis over the past decade. Because of the existence of multiple H-bonding and rigid heterocyclic ring of diphenylprolinols, they have mostly been used to catalyze the enantioselective alkylation,¹⁶ organocatalytic Michael addition,¹⁷ direct Aldol reaction¹⁸ and acyloin condensation,¹⁹ as well as promising applications, especially in efficient construction of axially chiral allenes or allenols.²⁰ However, to the best of our knowledge, there is no report on the use of structurally simple diphenylprolinols as CSAs for the analysis of chiral carboxylic acids. In recent years, we have paid continuing attention to develop new and effective CSAs for the determination of enantiomeric excess and the application of chiral discrimination.²¹ Therefore, in continuation of our work on the development of structurally simple and effective CSAs in this aspect, we herein report L-proline-based diphenylprolinols and their recognition ability for carboxylic acids by ¹H NMR spectroscopy.

The chiral *N*-benzyl-(*S*)-diphenyl(pyrrolidin-2-yl)methanol **1a** was synthesized starting from L-proline according to the literature procedure,²² and the subsequent hydrogenolysis of **1a** gave (*S*)-diphenyl(pyrrolidin-2-yl)methanol **1b** in good yield. Their structures are shown in Scheme 1.

In order to explore the binding properties of chiral diphenylprolinols (*S*)-**1a** and (*S*)-**1b**, they were then screened to see their efficacy in binding with a test sample of DL-mandelic acid in ¹H NMR analysis. The NMR experiments were performed with stoichiometric amounts of mandelic acid and CSA (1 : 1) (10 mM, in CDCl₃). The upfield change in the position of the signal of C^αH proton of mandelic acid upon treatment with CSA was measured ($\Delta\delta$) while the degree of splitting was measured by the differences in the separated peaks in terms of chemical shift non-equivalences as chemical shift change ($\Delta\Delta\delta$). The two entries of Table 1 indicate that the *N*-Bn, OH derivative CSA **1a** showed poor recognition (entry 1), while the basic unit (*S*)-CSA **1b** with free NH and OH groups showed very good recognition (entry 2). This observation perhaps indicates that the presence of the complex system is sufficient to offer multiple intermolecular hydrogen-bonding between the free NH and OH of the CSA and the test substrate. Therefore, the (*S*)-CSA **1b** was determined as the most effective receptor, and conducted a further study.

To further test the ability for enantiomeric discriminating of chiral diphenylprolinol **1b** as CSA for carboxylic acids, we first recorded ¹H NMR of 10 mM (*R*)-, (*S*)- and racemic mandelic acid (MA). The addition of (*S*)-CSA **1b** to a solution of racemic MA in CDCl₃ caused C^αH resonance in the ¹H NMR spectrum of MA to shift upfield and, in most cases, to split

Table 1 Effect of diphenylprolinols as CSAs on the α -proton of racemic mandelic acid. [$\Delta\delta$ = induced chemical shift; $\Delta\Delta\delta$ = chemical shift non-equivalences]

Entry	Diphenylprolinols	Probe signal PhCH(OH)COOH	
		$\Delta\delta^a$ (ppm)	$\Delta\Delta\delta$ (ppm)
1	(<i>S</i>)- 1a	−0.28	0.027
2	(<i>S</i>)- 1b	−0.38	0.062

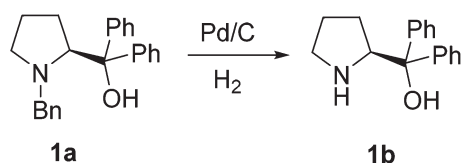
^aThe difference between the signals of DL-mandelic acid in CDCl₃ solution and the average of the signals of the two enantiomers after the addition of the CSA.

into two equal-intensity singlets that drift downfield slightly. Representative spectra for the most effective (*S*)-CSA **1b** are presented in Fig. 1a. The two singlets were assigned to the MA enantiomers, which become desymmetrized through their interaction with the diphenylprolinol.

The stoichiometry was determined according to the Job's method of continuous variation. The Job plots of $X \cdot \Delta\delta$ versus the molar fraction (*X*) of (*R*)- or (*S*)-mandelic acid in the mixture were obtained, which showed a maximum at *X* = 0.5, and the best baseline resolution was both achieved with a 1 : 1 ratio of (*S*)-**1b**/MA-2, as is shown in Fig. 1b. This indicated that (*S*)-**1b** and the acid bind in a 1 : 1 complex under these conditions.

To further explore the practical applications in the general ability of discrimination of (*S*)-CAS **1b** for a variety of analytes by ¹H NMR spectroscopy, the following carboxylic acids were chosen as guests (Fig. 2). These carboxylic acids vary from one another in their substituent group. The $\Delta\Delta\delta$ values of α -H signals were appropriate to give a good baseline resolution for most of the tested analytes which ranges from 0.005 to 0.096 ppm, the results are summarized in Table 2. The *p*-substituted aromatic hydroxy acids (Table 2, entries 2–6) almost showed a more bigger $\Delta\Delta\delta$ value compared with the *o*-substituted aromatic hydroxy acids (Table 2, entries 7–10). In particular, *p*-CF₃-substituted aromatic carboxylic acid ($\pm$)-**3** showed the biggest $\Delta\Delta\delta$ value as 0.076 ppm (Table 2, entry 2), while the *o*-halogen-substituted aromatic hydroxy acid showed the lowest $\Delta\Delta\delta$ value as 0.005 ppm (Table 2, entry 10). Similar results can be obtained for the *m*-substituted aromatic hydroxy acids (Table 2, entries 11 and 12), the above results indicated that the enantiodifferentiation of (*S*)-CAS **1b** could be possibly weakened due to steric hindrance of the *o*-substituted group. It is also interesting to note that chiral discrimination was observed for the OMe signals of ($\pm$)-**7** (Table 2, entry 6). The aromatic carboxylic acid with no α -OH chiral discrimination was also observed for the Me/OMe and methine proton (C^αH) signals (Table 2, entries 13 and 14).

What's more, (*S*)-CAS **1b** can even successfully discriminate the enantiomers of aliphatic carboxylic acids (Table 2, entries 15–19), especially for the pyromucic acid ($\pm$)-**16** showed the biggest $\Delta\Delta\delta$ value as 0.096 ppm (Table 2, entry 15), while the propionic acid derivatives ($\pm$)-**19–20** gave poor results (Table 2, entries 18 and 19).



Scheme 1 The structures of chiral diphenylprolinols, (*S*)-**1a** and (*S*)-**1b**.

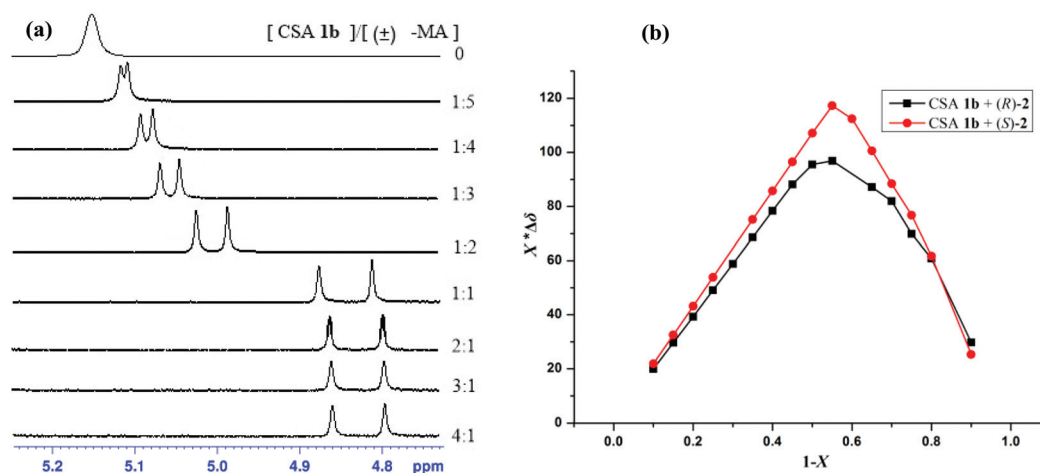


Fig. 1 (a) Variations in part of the 500 MHz ^1H NMR spectrum corresponding to the C^αH resonance (*rac*)-mandelic acid (MA, 10 mM in CDCl_3) upon increasing (*S*)-CSA **1b**; (b) Job plots of (*S*)-CSA **1b** with (*R*)- and (*S*)-**2**. $\Delta\delta$ stands for the chemical shift change of the $\alpha\text{-H}$ proton of (*R*)- and (*S*)-**2** in the presence of (*S*)-CSA **1b**. X stands for the molar fraction of the (*S*)-CSA **1b** ($X = [(\text{S})\text{-CSA } \mathbf{1b}]/[(\text{S})\text{-CSA } \mathbf{1b}] + [\mathbf{2}]$). The total concentration is 10 mM in CDCl_3 .

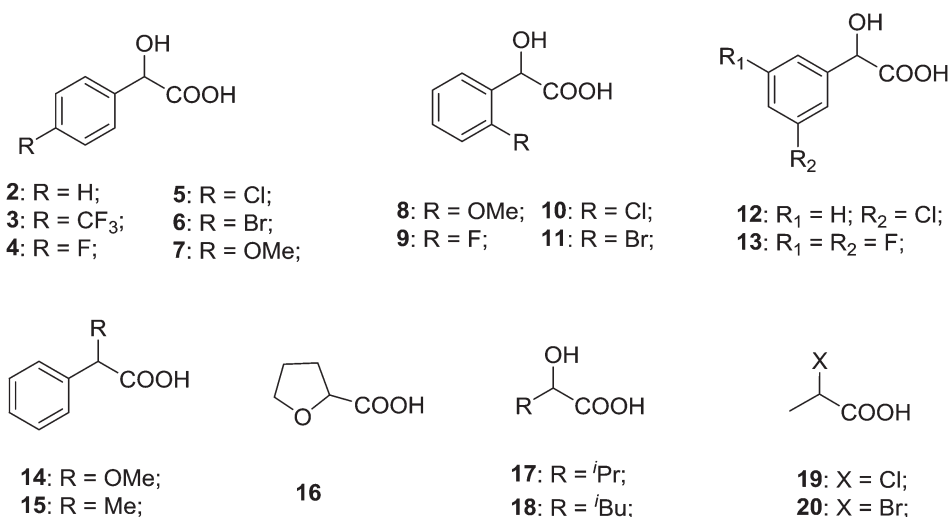


Fig. 2 Structures of the tested *rac*-carboxylic acids.

To explore the quantitative analysis ability of (*S*)-**1b** as a CSA for enantiomeric determination, the ee values of non-racemic mandelic acid **7** samples were determined by integration of the $\alpha\text{-H}$ signal of **7** in ^1H NMR. The results shown in Fig. 3 were calculated based on the integrations of the NMR signals, and were within $\pm 1\%$ of the actual enantiopurity of the samples. The linear relationship between the NMR-determined values and those gravimetry determined values is excellent with $R^2 = 0.9995$.

Having demonstrated enantiodiscrimination of the above carboxylic acids, we then tried to investigate the applicability of CSA **1b** for the determination of absolute configuration of chiral carboxylic acids by a suitable method.^{14d,23} The NMR spectral behavior of a series of chiral carboxylic acids of known absolute configuration was studied using (*S*)-CSA **1b** and its

enantiomer (*R*)-CSA **1b** to find the existence of any correlation between the absolute configuration and the NMR chemical shifts (Table 3). The results indicate that negative $\Delta\delta_{\alpha\text{-H}}^{R,S}$ correlates with (*S*)-carboxylic acids and positive $\Delta\delta_{\alpha\text{-H}}^{R,S}$ correlates with (*R*)-carboxylic acids.

We have developed structurally simple diphenylprolinols and tested their efficacy as CSAs in ^1H NMR spectroscopy to discriminate carboxylic acids. A linear relationship has been observed between the experimental and observed values of ee indicating the possible use of these compounds for quick and reliable analysis of enantiomerically enriched samples of mandelic acid. From the experiments performed a preliminary conclusion indicated that the diphenylprolinol **1b** with the free NH and OH is most effective in the chiral discrimination of carboxylic acids in ^1H NMR.

Table 2 Measurements of ^1H chemical shift non-equivalence ($\Delta\Delta\delta$) values of racemic carboxylic acids in the presence of (*S*)-CSA **1b** by ^1H NMR spectroscopy in CDCl_3 at 25 $^\circ\text{C}$ ^a

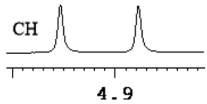
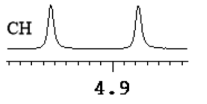
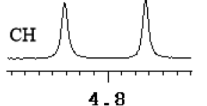
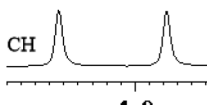
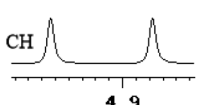
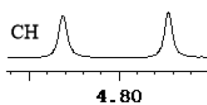
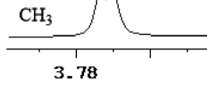
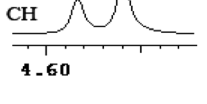
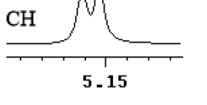
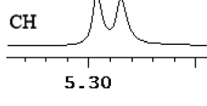
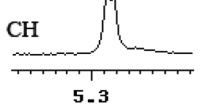
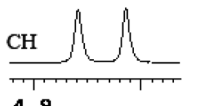
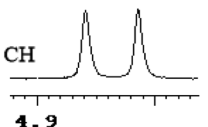
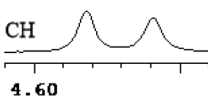
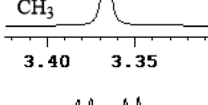
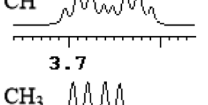
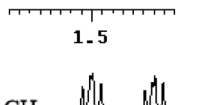
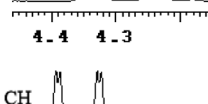
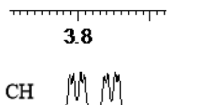
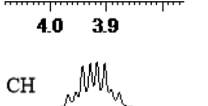
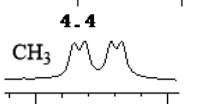
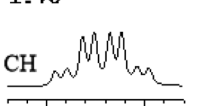
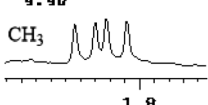
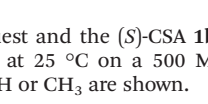
Entry	Carboxylic acids	$\Delta\Delta\delta^b$ (ppm)	$\Delta\Delta\delta^b$ (Hz)	Spectrum ^b
1	(±)-2	0.062	31	
2	(±)-3	0.076	38	
3	(±)-4	0.056	28	
4	(±)-5	0.058	29	
5	(±)-6	0.058	29	
6	(±)-7	0.060	30	
		0.003	1.5	
7	(±)-8	0.022	11	
8	(±)-9	0.008	4	
9	(±)-10	0.012	6	
10	(±)-11	0.005	2.5	
11	(±)-12	0.044	22	

Table 2 (Contd.)

Entry	Carboxylic acids	$\Delta\Delta\delta^b$ (ppm)	$\Delta\Delta\delta^b$ (Hz)	Spectrum ^b
12	(±)-13	0.050	25	
13	(±)-14	0.023	11.5	
		0.003	1.5	
14	(±)-15	0.052	26	
		0.014	7	
15	(±)-16	0.096	48	
16	(±)-17	0.057	28.5	
17	(±)-18	0.061	30.5	
18	(±)-19	0.007	3.5	
		0.003	1.5	
19	(±)-20	0.006	3	
		0.014	7	

^a Typical conditions: concentration of the guest and the (*S*)-CSA **1b** is 10 mM (1 : 1) and the spectra are recorded at 25 $^\circ\text{C}$ on a 500 MHz spectrometer. ^b $\Delta\Delta\delta$ values and spectra for $\alpha\text{-H}$ or CH_3 are shown.

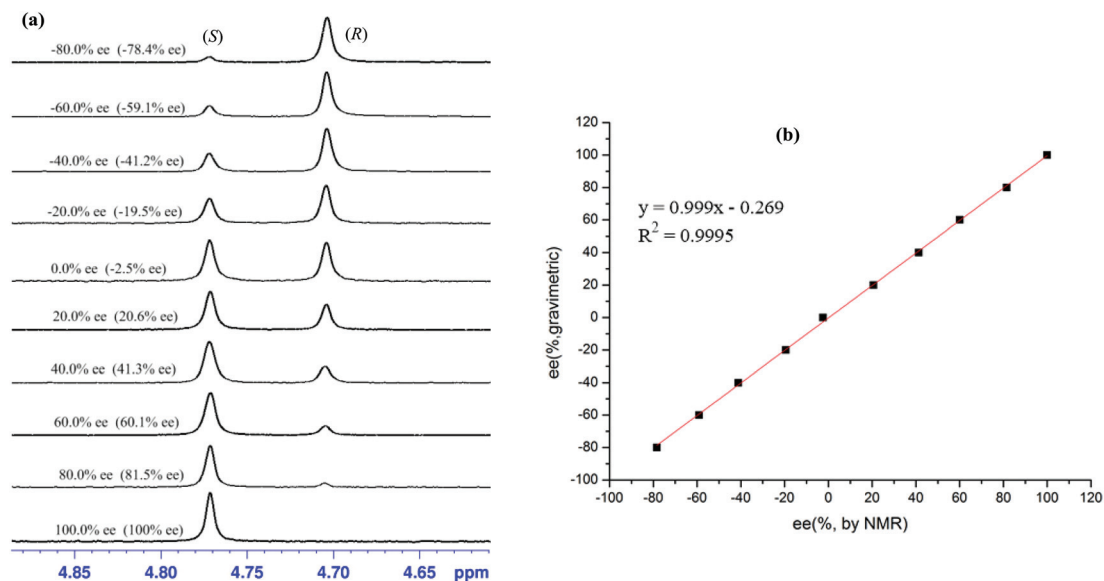


Fig. 3 (a) Selected regions of the ^1H NMR spectra of non-racemic **7** samples (varied ee values) with (S)-CSA **1b** in CDCl_3 ; (b) linear correlation between ee values determined by gravimetry and NMR ee values is defined in terms of (S)-**7**, R^2 = correlation coefficient.

Table 3 Measurements of $\Delta\delta_{\alpha\text{-H}}^{R,S}$ of chiral carboxylic acids in the presence of (R)-CSA **1b** and (S)-CSA **1b**^a

Entry	Chiral carboxylic acids	$\Delta\delta_{\alpha\text{-H}}^{R,S}$ ^b (ppm)
1	(R)-Mandelic acid	+0.026(R)
2	(S)-Mandelic acid	-0.027(S)
3	(R)-4-Chloromandelic acid	+0.022(R)
4	(S)-4-Chloromandelic acid	-0.030(S)
5	(R)-4-Methoxymandelic acid	+0.027(R)
6	(S)-4-Methoxymandelic acid	-0.031(S)
7	(R)-3-Chloromandelic acid	+0.014(R)
8	(R)-2-Phenylpropionic acid	+0.006(R)
9	(S)-2-Tetrahydrofuroic acid	-0.163(S)
10	(S)-2-Hydroxy-3-methylbutanoic acid	-0.069(S)
11	(S)-2-Hydroxy-4-methylpentanoic acid	-0.070(S)
12	(S)-2-Hydroxy-3,3-dimethylbutyric acid	-0.149(S)

^a $\Delta\delta_{\alpha\text{-H}}^{R,S}(\Delta\delta^R - \Delta\delta^S)$ values for $\alpha\text{-H}$ chiral carboxylic acids are shown.

^b Assigned configurations are labeled in parentheses.

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